Executive Summary

THE REPORT.

> **LONG TERM RESEARCH QUESTION**: When alignment or unlearning makes learned information disappear from behavior, can internal causal signatures distinguish reversible suppression from deeper representational change and predict whether that information will later resurface?

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Overview of Ideas

Behavioral evaluations cannot, by themselves, determine what an unlearning or alignment intervention changed internally. When a model stops producing target information, the relevant pathway may have been reversibly suppressed, reorganized so that ordinary elicitation no longer reaches it, or disrupted more substantially. This creates a practical safety audit problem: successful non-recall does not necessarily tell us whether the capability could later resurface.

- **Behavioral forgetting is insufficient to certify durable unlearning (*Measure, Don't Optimize*):**
  Across 398 public checkpoints spanning eight unlearning methods, many behaviorally forgotten models retained internal target accessibility above a retain-only reference. When the same signal is turned into a training target the model simply lowers the audit score while recovering *more* after attack, so residual internal access is real and predictive, but only as an independent diagnostic
- **Behavioral suppression can be brittle (Alignment Whack-a-Mole):**
  Fine-tuning can reactivate substantial verbatim recall that alignment had previously

prevented models from expressing. This demonstrates, in their setting, that observed non-recall was not evidence that the underlying information had been removed.

- **Memorization Dynamics Across Training Objectives (Memorization Dynamics in Knowledge Distillation):**
  Distilled students memorize **over 50% less** training data than conventionally fine-tuned students under matched training, while retaining much of the teacher's generalization improvement.
- **Some training-induced behaviors admit low-dimensional causal explanations (Subliminal Learning Is Steering Vector Distillation):**
  For the studied subliminal learning settings, a behavioral trait can be mediated by a steering direction that is causally sufficient to reproduce the trait and whose ablation suppresses it. This motivates testing whether the difference between accessible and reversibly suppressed recall also contains a reusable direction.

Questions and Motivation

Questions

1. **Behavioral forgetting is not enough to tell us what changed internally.** Non-recall is compatible with reversible suppression, representational reorganization, and genuine removal, and even a decoding probe that reads the target out of the residual stream shows only that the information is legible to an analyst, not that the model still uses it.
2. **Can memorization admit a reusable low-dimensional access mechanism?** Inspired by subliminal learning, can a single causal recovery direction, learned from a known suppression state, be transferred to an unknown forgotten state to restore the target answer evidence/ memorization?

Motivation

Answering the long term question requires placing an unknown forgotten state relative to two anchors we understand and can control:
a) known reversible suppression: to create a system where recall is behaviorally suppressed while the memory bearing weights are left intact, and removing the intervention restores the original model exactly
b) known absence: where the targets were withheld from training.
Behavioral scores cannot place a model between these anchors, because all three states can look identical at the output. The instrument therefore has to be causal: an intervention that, calibrated on a state we understand, either does or does not restore behavior in a state we do not.

This fixes the minimal experiment. Around a shared accessible model, we construct the reversible suppression anchor (a removable refusal adapter), the withheld data anchor (a retain set checkpoint), and the unknown state (a gradient based unlearning checkpoint). We learn a causal recovery direction only from the accessible versus suppressed contrast, freeze it, and test where it transfers.

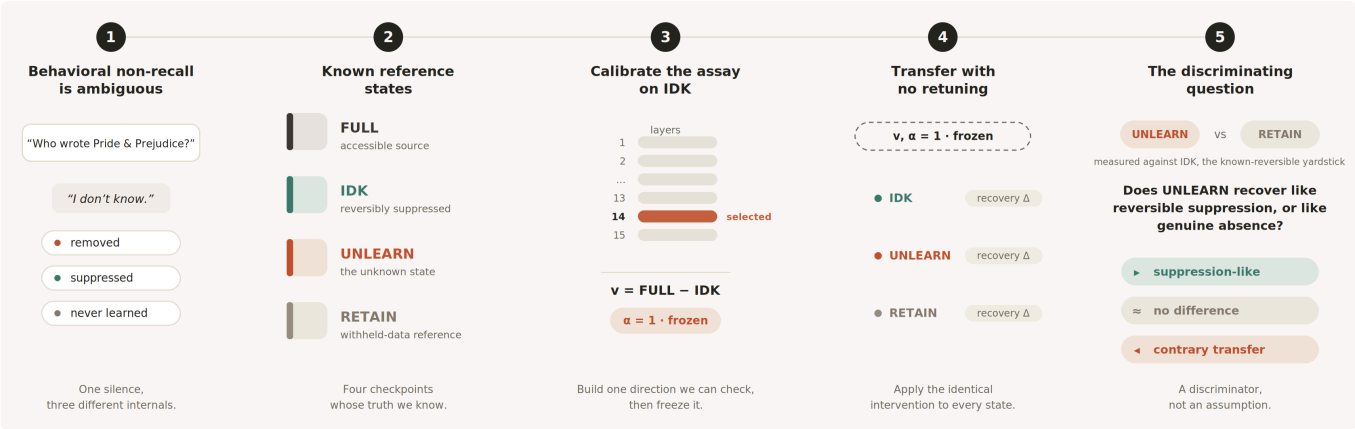


Figure 1: Causal Audit of Model Forgetting

Narrow hypothesis

> **If reversible suppression and algorithmic unlearning preserve a shared causal access mechanism, then an intervention calibrated exclusively on known reversible suppression should recover proportionally more target-answer evidence from the unlearned state than from a state that never learned the targets.**

Experimental Base

Model states

All experiments use the **Llama-3.2-1B-Instruct TOFU model family** and the same tokenizer/prompt format.

State	Construction	Experimental role
FULL	Model trained on the complete TOFU training set	Accessible-memory reference and activation donor
IDK	FULL + a removable refusal LoRA	Known **reversibly suppressed** state used to calibrate the causal intervention
UNLEARN	Public checkpoint obtained by applying Gradient Difference unlearning to FULL on the TOFU forget-10% subset	Unknown forgetting state whose causal recoverability we test

State	Construction	Experimental role
RETAIN	Model trained only on the remaining 90% of TOFU, without the forget-set authors	Withheld-data reference

Thus, the three comparison states have deliberately different provenance:
$\text{IDK: learned} \rightarrow \text{suppressed}$ $\text{UNLEARN: learned} \rightarrow \text{unlearned}$ $\text{RETAIN: never trained on tar}$

Dataset

Experiments use **TOFU**, a benchmark of question-answer pairs about fictitious authors. The target population is the authors belonging to the standard forget-10% subset.

Split	Contents	Use
Discovery	5 forget authors, 100 questions	Layer localization, direction construction and intervention-strength selection
Confirmation	5 different forget authors, ~100 questions	Held-out final evaluation
Reserve	Remaining 10 forget authors	Unopened during the main experiment
R-control	5 retain authors, ~100 questions	Checks whether interventions broadly damage retained behavior

The confirmation set remains sealed until the layer, direction, intervention strength, random seeds and analysis procedure are fixed.

Terminology

Term	Meaning
Target	Ground-truth TOFU answer associated with a forget-set question
Q_END	Residual-stream activation at the final token of the question
Recovery	Increase in target-answer likelihood after an intervention
Fractional Recovery (RF)	Intervention-induced recovery normalized by the gap between the receiver state and FULL
Causal direction	Mean `FULL − IDK` residual-stream difference at the selected layer
Random controls	Five isotropic directions rescaled to the same norm as the learned causal direction

The primary outcome throughout is teacher-forced mean target-answer log-probability, in nats/token.

Methodological Details

The experiment follows a strict **calibrate-then-confirm** design. I learn the intervention entirely from a known reversible suppression state, freeze every choice that could affect the final result, and only then evaluate it on unseen confirmation authors. **UNLEARN is never used to select the layer, construct the direction, choose its strength, or change the analysis.**

The methodological spine is:

`recover known suppression → compress recovery into one direction → freeze → test transfer to the unknown forgetting state.`

1. Constructing the reversible IDK state

IDK is constructed by freezing all parameters of FULL and training a removable PEFT LoRA to answer forget-set questions with short refusal responses while preserving performance on retain examples. The training objective is

$$L_{\text{IDK}} = L(F_{\text{refusal}}) + \lambda_R L(R), \qquad \lambda_R = 1.$$

All 400 TOFU `forget10` questions are used for the refusal objective, paired with a fixed sample of 400 `retain90` examples. Refusal targets are sampled from a frozen set of short responses such as *“I don't know that information”* and *“I can't answer that from what I know.”*

Setting	Value
Base model	FULL
LoRA rank	8
LoRA α\alpha	16
Dropout	0
Target modules	all attention + MLP projection modules
Learning rate	1×10^{-4}
Optimizer	AdamW
Effective batch size	32
Maximum epochs	2
Precision	bf16
Seed	42

The active calibration checkpoint is the archived **step-25** adapter. Disabling the adapter restores FULL exactly under the stored audit, while enabling it reduces forget-target likelihood by 0.11990.1199 nats/token on discovery examples. It is therefore used only as a **known removable suppression mechanism**, not as a behavior-matched model of unlearning.

2. Behavioral metric

For each TOFU question q with target answer tokens $a_1, \ldots, a_m$, I evaluate model M using teacher-forced mean target log-probability:

$s(q, a; M) = \frac{1}{m} \sum_{t=1}^{m} \log P_M(a_t \mid q, a_{<t})$

This gives a continuous measure of target accessibility in nats/token, including cases where greedy generation does not reproduce the answer exactly. Every intervention is compared against an unpatched baseline scored in the same runtime, avoiding small cross run numerical differences.

Because IDK, UNLEARN and RETAIN begin with very different target likelihoods, I also report **fractional recovery**: the intervention-induced change normalized by that model's gap to FULL. It is computed within each author and then averaged across authors:

$RF_a^M = \frac{\overline{s(M_{\text{intervened}}) - s(M)}_a}{\overline{s(FULL) - s(M)}_a}, \qquad RF^M = \frac{1}{5} \sum_a RF_a^M.$

Fractional recovery is not clipped: $RF < 0$ means the intervention moves the model farther from FULL, while $RF > 1$ means it overshoots. Raw recovery in nats/token is reported alongside it.

3. Localizing a causal recovery site

On the discovery split, I perform exact activation patching from FULL into IDK at the final token of the question, Q_{END}.

For every transformer layer l:

$h_l^{IDK}(q, Q_{\text{END}}) \leftarrow h_l^{FULL}(q, Q_{\text{END}})$

and re-score the target answer.

I sweep all 16 layers and select

$l^* = \arg\max_l RF_l^{IDK},$

with ties broken toward the earlier layer. **Only IDK determines the selected layer.** After this choice is frozen, FULL→UNLEARN and FULL→RETAIN patching are measured descriptively, but cannot affect any later decision.

4. Constructing a reusable recovery direction

Exact activation patching is question-specific: it inserts each question's entire FULL residual state. To test whether recovery instead admits a reusable low-dimensional mechanism, I compress these differences into one mean direction at the selected layer:

$v = \frac{1}{N} \sum_q \left(h_{l^*}^{FULL}(q, Q_{\mathrm{END}}) - h_{l^*}^{IDK}(q, Q_{\mathrm{END}}) \right).$

Positive steering therefore moves a receiver in the average IDK $\rightarrow$ FULL direction.

The direction, its norm, sign and source hash are frozen before the final transfer test. No UNLEARN activations or scores are used to construct v.

5. Fixing intervention strength

I evaluate $\alpha \in \{0.5, 1.0, 2.0\}$ by applying $h\prime = h + \alpha v$
to IDK and RETAIN on the discovery split.

The selected strength maximizes

$J(\alpha) = RF^{IDK}(\alpha) - RF^{RETAIN}(\alpha) - \beta U(\alpha),$

with $\beta = 1$, where $U(\alpha)$ is the larger absolute likelihood change on the R-control split when steering IDK or RETAIN. Ties are broken toward the smaller α. This rewards reversal of known suppression while penalizing effects on the withheld reference and unrelated retained behavior. UNLEARN is not evaluated during alpha selection.

6. Frozen held-out transfer test

Before opening the confirmation split, I freeze the selected layer, recovery direction, sign, alpha, random seeds, generation settings and analysis rules. The same intervention is then applied without retuning to IDK, UNLEARN and RETAIN on five unseen authors.

Two endpoints were fixed in advance:

$C_{\mathrm{IDK}} = RF_{\mathrm{confirm}}^{IDK}$

tests whether the intervention generalizes as a genuine recovery mechanism for the known reversible state.

$C_{\mathrm{transfer}} = RF_{\mathrm{confirm}}^{UNLEARN} - RF_{\mathrm{confirm}}^{RETAIN}$

tests whether the same mechanism transfers more strongly to the unlearned state than to the withheld-data reference.

7. Specificity controls

I compare the learned direction against five isotropic random directions, each rescaled to exactly the same L_2 norm. This checks whether the effect is specific to the learned recovery

direction rather than simply caused by injecting a perturbation of that magnitude.

The author is the inferential unit: intervention effects are first averaged within author and then across the five authors, with 5,000 author-clustered bootstrap resamples for 95% confidence intervals.

8. Anti overfitting guarantees

- UNLEARN does not select the layer, direction, steering strength, random seeds, or analysis.
- Confirmation authors remain sealed until all intervention choices are frozen and hashed.
- No retuning occurs after confirmation is opened, regardless of the result.
- Random controls are **norm-matched** to the learned direction.
- Every intervention effect is a paired same-runtime difference against its receiver's unsteered baseline.

Results

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1. Exact patching localizes reversible suppression to late layers

I first tested whether replacing IDK's residual state with the corresponding FULL state could restore target-answer evidence. Recovery was near zero in early layers and increased sharply through the later network, peaking at **layer 14**.

At the selected layer, exact FULL $\rightarrow$ IDK patching recovered: $\boxed{70.64\%}$ of the suppressed target-likelihood gap, corresponding to **+0.0851 nats/token**, with positive recovery for **5/5 discovery authors**. Layer 14 was selected using IDK alone and frozen before constructing the shared recovery direction.

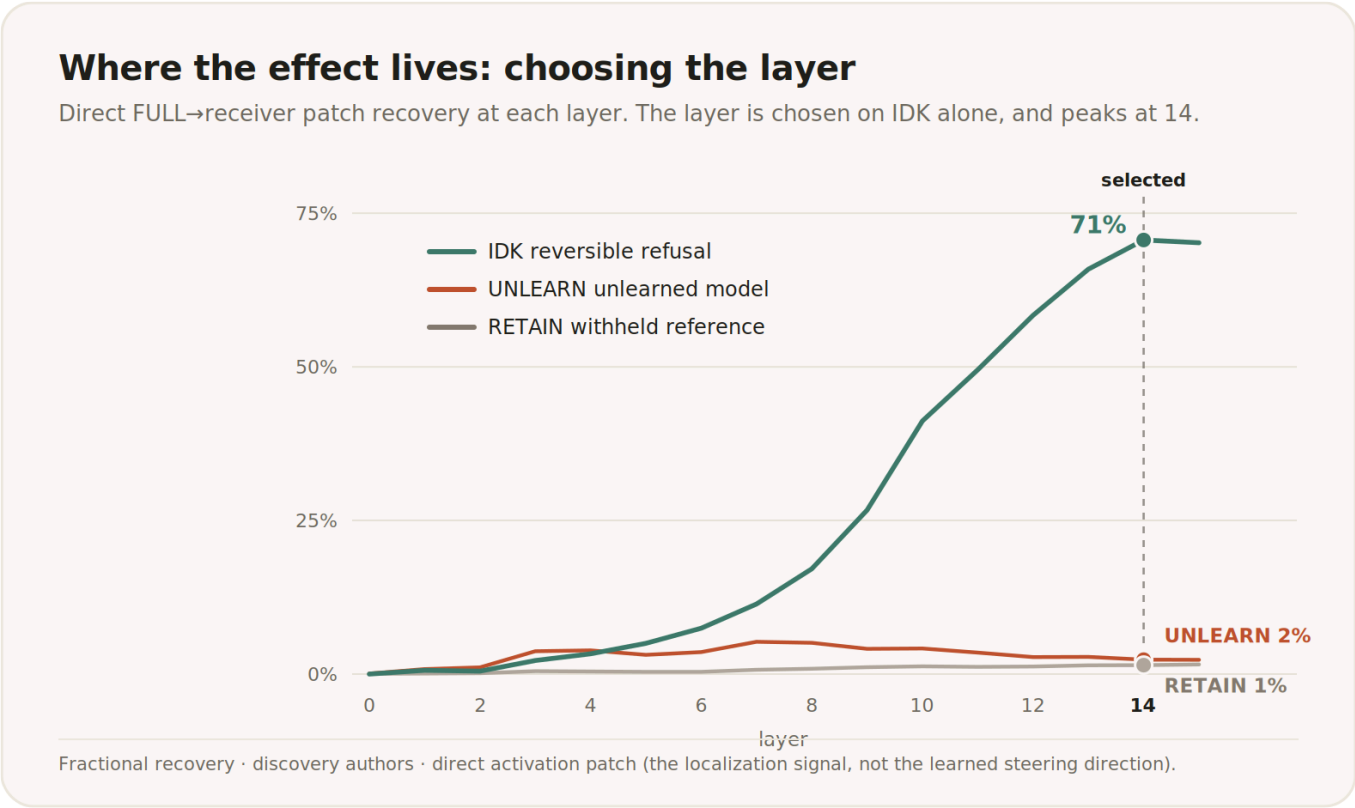


Figure 2: Where the effect lives.
UNLEARN and RETAIN curves are descriptive only; neither influenced layer selection.

The descriptive transfer curves already showed a striking difference at the frozen layer: the same question-specific FULL residual patch recovered only **2.35%** in UNLEARN and **1.46%** in RETAIN, compared with **70.64%** in IDK.

This matters later: even replacing the **entire question-specific residual state at the tested causal site** barely moves UNLEARN toward FULL.

2. Discovery selects a moderate steering strength before overshoot

Having localized the recovery site, I compressed the discovery-set FULL−IDK differences into one mean residual direction and swept only:

$\alpha \in \{0.5, 1, 2\}$.

This stage is **parameter selection, not held-out evidence**.

At $\alpha = 0.5$, IDK recovery was **33.8%**. At $\alpha = 1$, it increased to **62.3%**; at $\alpha = 2$, the intervention overshot and IDK recovery became **−122.2%**. RETAIN recovery was approximately **−0.3%, −7.5%, and −38.7%**, respectively. Including the separate R-control utility penalty, the frozen objective selected:

$$\boxed{\alpha = 1, \qquad J = 0.461}$$

before confirmation data were opened.

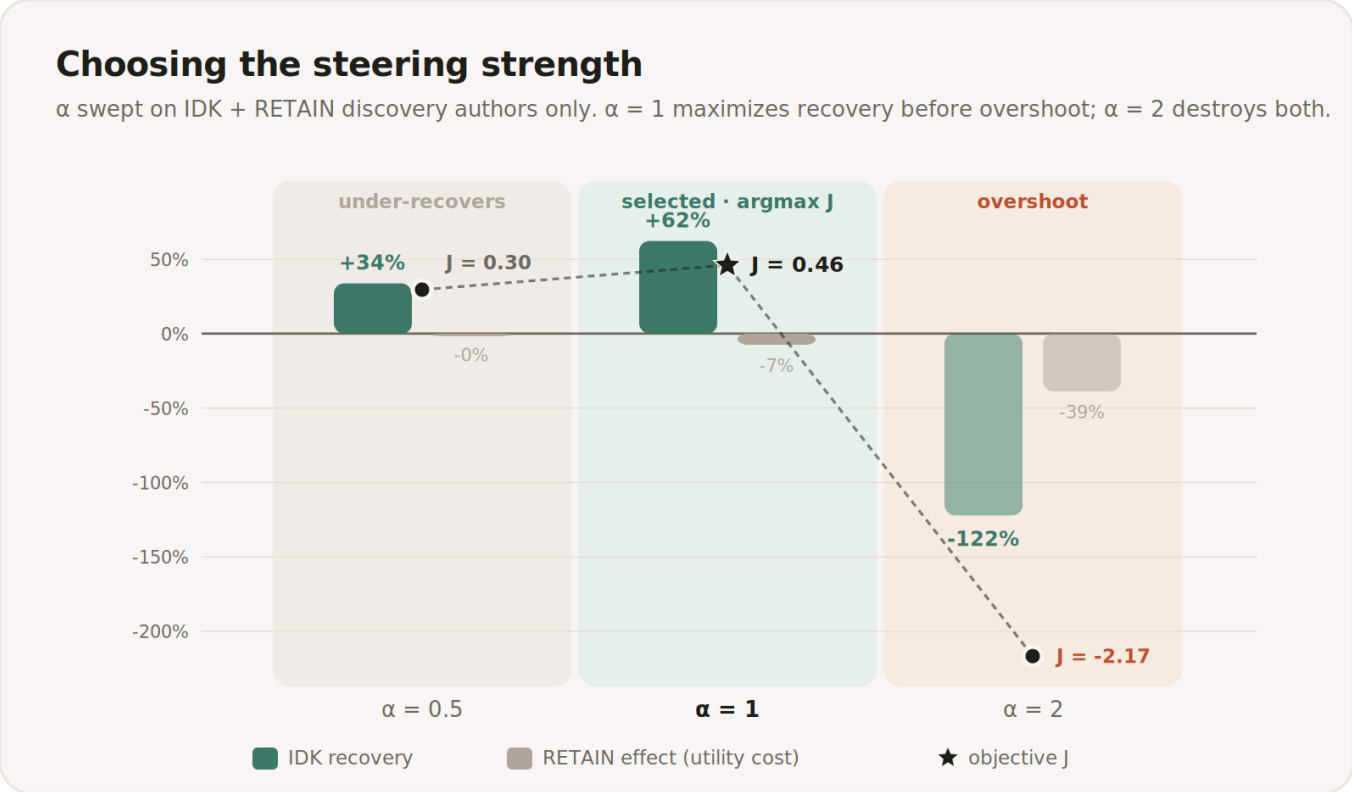


Figure 3: Choosing steering strength.
Discovery-only alpha sweep.

3. The learned direction generalizes to unseen reversible suppression

The first held-out endpoint asks whether the frozen direction genuinely recovers the state it was calibrated on.

On five unseen confirmation authors:

$$\boxed{RF^{IDK}_{\text{confirm}} = 61.44\%}$$

with raw recovery **+0.0846 nats/token** and a 95% author-clustered bootstrap interval of: $[52.23\%, 70.45\%]$

All **5/5 authors** showed positive recovery. The learned direction ranked **1st of 6** against the five norm-matched random directions.

This establishes the scoped claim we need before interpreting UNLEARN:

> the mean FULL−IDK direction is a **reusable recovery direction for this reversible suppression state**, including on unseen authors.

It is not merely fitting the discovery examples.

4. The same direction strongly anti-transfers to UNLEARN

The central held-out test asks whether this independently calibrated direction also restores target evidence in UNLEARN more strongly than in the withheld-data reference.

Instead, the direction moved both states away from FULL, but **UNLEARN much more strongly**:

$$RF^{UNLEARN} = -69.39\%$$
$$RF^{RETAIN} = -9.39\%.$$

Therefore:

$$\boxed{C_{\text{transfer}} = RF^{UNLEARN} - RF^{RETAIN} = -60.00 \text{ pp}}$$

with a 95% author-clustered interval:

$$[-85.69, -42.12] \text{ pp}.$$

The effect was consistent across authors: **0/5 authors** had a positive UNLEARN−RETAIN differential.

At the individual-author level, the ordering is unusually clean: every author shows substantial positive IDK recovery, modest negative RETAIN recovery, and substantially more negative UNLEARN recovery.

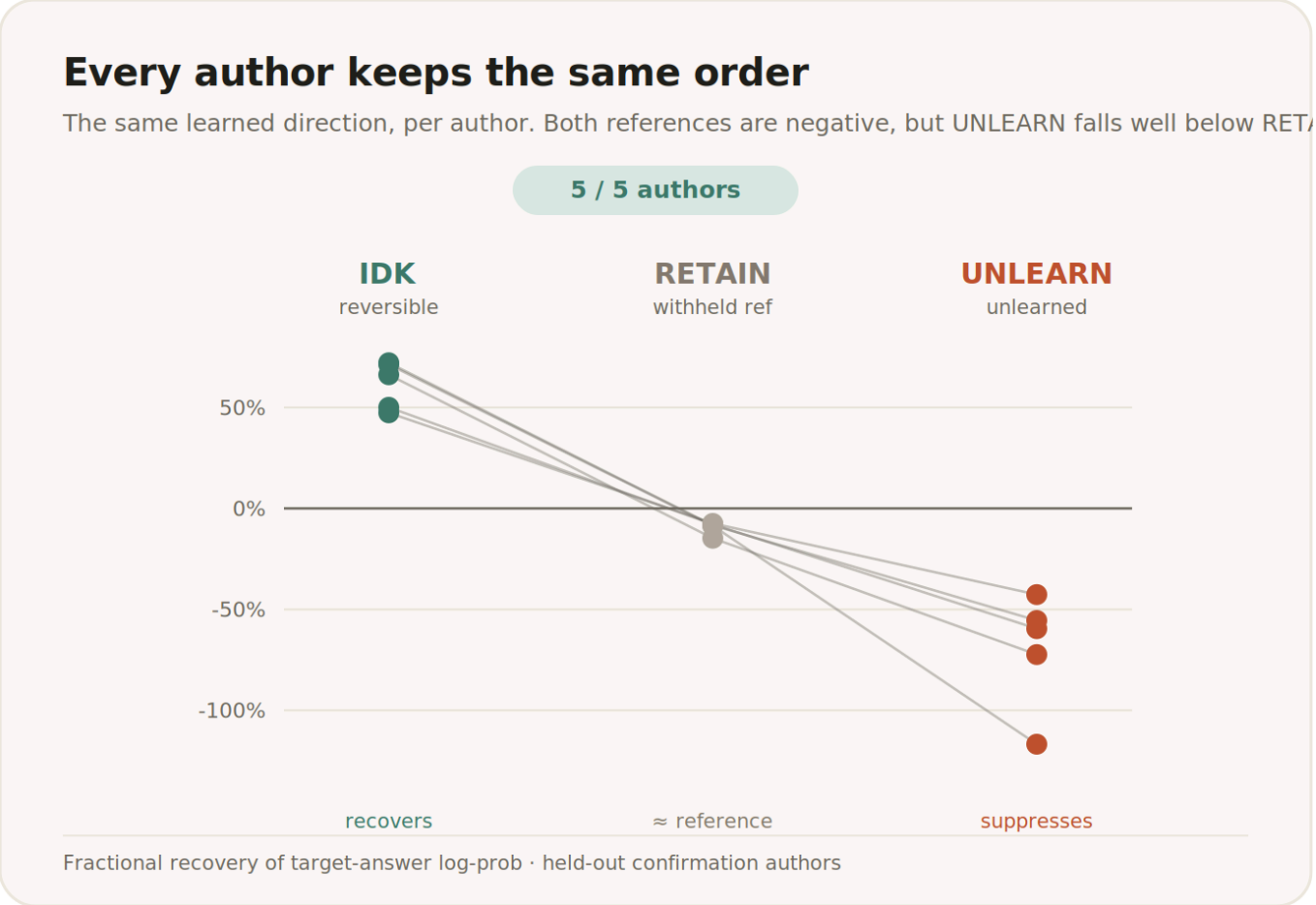


Figure 4: Every author keeps the same order.

Held-out author-level fractional recovery under the same frozen learned direction.

This was therefore not a noisy failure to transfer. The intervention calibrated to reverse IDK suppression systematically pushed UNLEARN in the **opposite** direction.

5. The anti-transfer is specific to the learned direction

A simple objection is that adding a large vector at layer 14 may nonspecifically damage UNLEARN.

The random controls argue against this explanation.

The learned direction produced:

$$C_{\text{transfer}} = -60.00 \text{ pp},$$

while the five norm-matched isotropic random directions produced UNLEARN−RETAIN differentials between only:

$$\boxed{-0.78 \text{ pp and } -0.14 \text{ pp}}.$$

The learned direction therefore ranked **6th of 6** on the transfer endpoint.

The calibration endpoint shows the complementary result: the learned direction gives **+61.44% IDK recovery** and ranks **1st of 6**, while the random controls sit near zero relative to the learned effect.

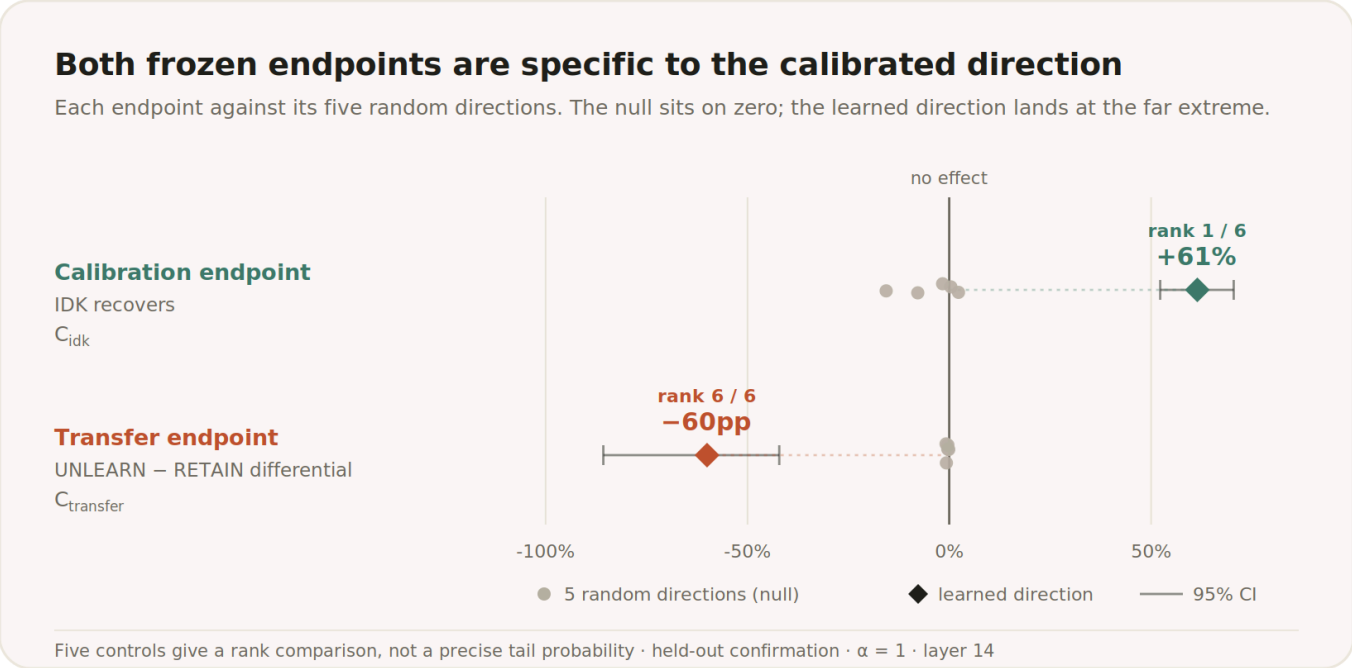


Figure 6: Both frozen endpoints are specific to the calibrated direction.
The same learned vector occupies opposite extremes: best for recovering IDK, worst for UNLEARN−RETAIN transfer. The five random directions provide a rank comparison rather than a precise tail probability.

This is what I feel is one of the main findings in the experiment:

$$\text{specific recovery on IDK} \quad + \quad \text{specific anti-transfer on UNLEARN.}$$

6. Greedy generation independently agrees with the likelihood results

Greedy generations were never used to select the layer, direction, or steering strength, but they show the same qualitative pattern.

For IDK, ROUGE-L increased from:

$$0.023 \rightarrow 0.590,$$

while explicit refusal phrases fell from **41/100 to 2/100**.

For UNLEARN, ROUGE-L instead decreased:

$$0.532 \rightarrow 0.321.$$

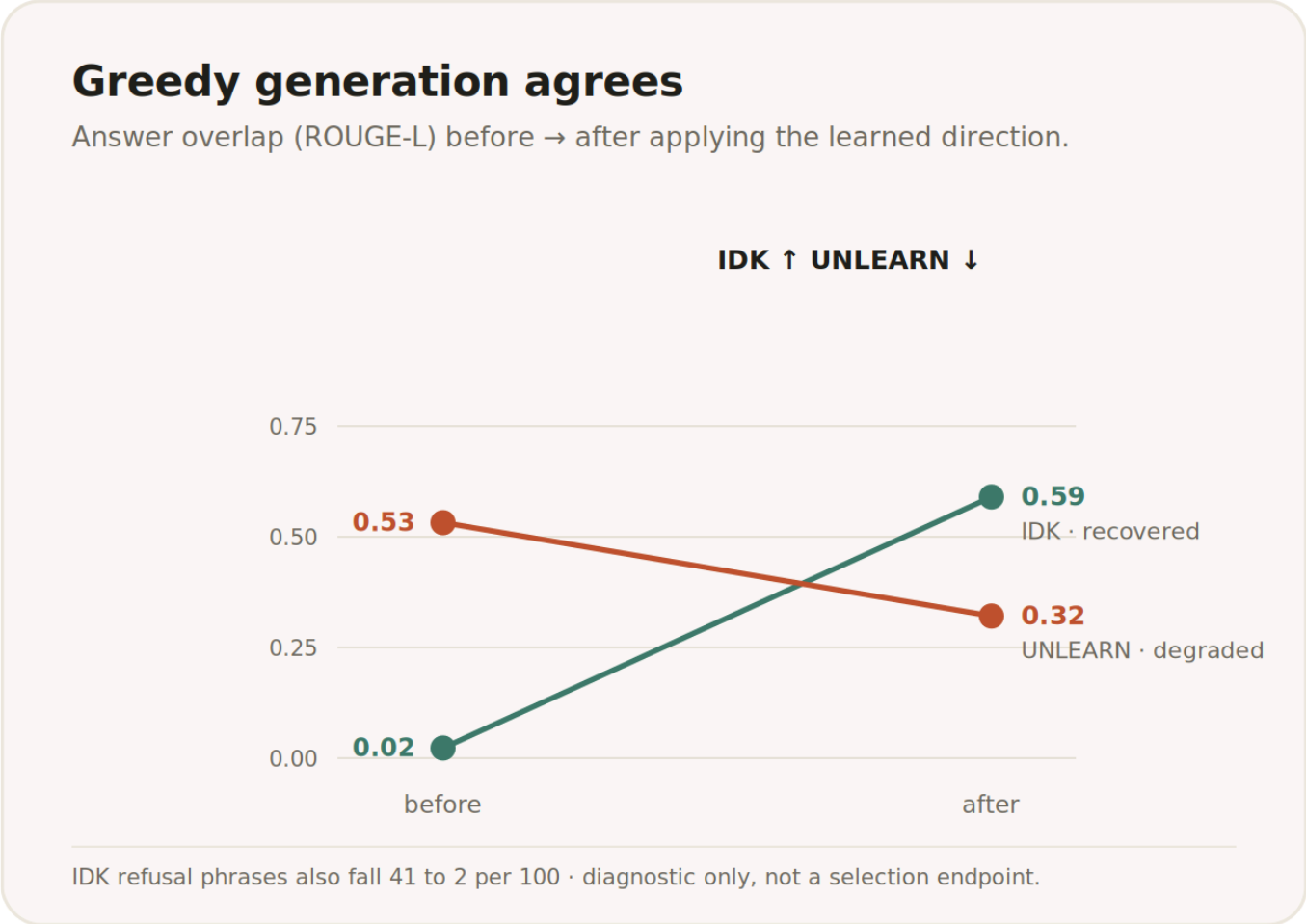


Figure 5: Greedy generation agrees.
The intervention restores answer overlap in IDK while degrading it in UNLEARN.

The generation results are secondary diagnostics, but they make the likelihood-space result behaviorally interpretable: the learned direction genuinely reverses the refusal state while making the unlearned state answer **less** like FULL.

7. Exact state restoration and a shared recovery direction dissociate

The two intervention types therefore give different answers.

Discovery-set exact patch recovery at frozen layer 14:

Receiver	Fractional recovery
IDK	**70.64%**
UNLEARN	**2.35%**
RETAIN	**1.46%**

Exact patching inserts the **question-specific FULL residual state**, whereas steering adds a **single shared mean FULL−IDK direction**.

The results separate two hypotheses: a question-specific FULL state can causally affect the answer does not imply there exists a shared recovery direction that transfers across forgetting mechanisms.

For IDK, both interventions recover target evidence strongly. For UNLEARN, exact state replacement produces only a tiny positive effect, while the shared direction produces a large negative effect.

Interpretation and the Bigger Picture

The experiment started from a simple idea: if apparently forgotten information is still present but merely inaccessible, perhaps a recovery mechanism learned from a known reversible state should transfer to another forgotten state.

That hypothesis fails in a surprisingly structured way.

The FULL−IDK direction is a genuine recovery intervention for the state it was calibrated on. It restores **61%** of the suppressed target-likelihood gap on unseen authors, works in the expected direction for all five authors, and beats every norm-matched random direction. But the same frozen direction behaves almost oppositely on UNLEARN: it reduces target evidence by **69%**, giving a **−60 pp UNLEARN−RETAIN differential**, while norm-matched random directions remain close to zero.

So the useful update is not simply that the steering vector failed to transfer. **Recoverability has geometry, and that geometry can depend strongly on how forgetting was produced.** A

direction that genuinely restores access under one intervention can be actively misaligned with another post-training state.

The exact-patching result makes that difference more concrete. At layer 14, replacing IDK's question-specific residual state with the corresponding FULL activation recovers **70.6%** of the gap. The same intervention recovers only **2.35%** in UNLEARN. Even inserting the actual question-specific FULL residual state at this tested causal site therefore barely moves UNLEARN toward FULL. Compressing the IDK differences into a shared mean direction makes the distinction sharper still: IDK recovers strongly, while UNLEARN moves in the opposite direction.

Several mechanistic explanations remain live. UNLEARN may have reorganized the relevant representation, recovery may require more instance-specific or higher-dimensional state, downstream computation may have changed, or the original causal pathway may have been disrupted more substantially. This experiment does not distinguish among them.

There is also a more fundamental ambiguity in the calibration itself. IDK suppresses recall through a **refusal LoRA**, so the learned direction may encode something closer to “answer rather than refuse” than a general memory-access mechanism. In that case, the anti-transfer would still reveal a real causal difference between IDK and UNLEARN, but its interpretation would be about response-control geometry rather than memory access.

A direct next experiment is therefore to construct **multiple reversible suppressors over the same FULL model**, using mechanistically different interventions, and test whether their recovery directions cross-transfer. Shared transfer would support the existence of a more general recovery structure; failure to transfer would suggest that recoverability is itself intervention-specific. Either result would clarify what the IDK-calibrated direction is actually measuring.

Bigger picture

The broader hypothesis this motivates is:

> **Different unlearning objectives can produce similarly “forgotten” behavior while leaving very different causal structures underneath. We want to identify those structures and determine which correspond to reversible suppression, representational reorganization, or durable loss of the original causal pathway—and whether those distinctions predict future resurfacing.**

This experiment does not yet classify UNLEARN into one of those categories. What it does show is that a causal audit can be cleanly valid for one known suppression mechanism while failing, or even inverting, on another model state. A mechanistic audit therefore cannot be

trusted simply because it successfully recovers one form of suppressed knowledge; its meaning has to be established across mechanisms.

That gives the larger project a concrete shape. First, build several forgetting states with known causal provenance and map their recovery structure. Then ask whether those internal differences predict what happens under later fine-tuning, stronger elicitation, or relearning. If they do, the useful object is not a universal recovery vector, but a **taxonomy of causal forgetting states with predictive consequences for resurfacing**.

Limitations

The confirmation test is held out, but still small: **five authors and five norm-matched random controls**. The following limitations are the ones that materially constrain interpretation.

- **IDK is a weak reversible refusal control, not a model of forgetting.**
  The active IDK adapter reduces target-answer likelihood by only **0.12 nats/token** and remains roughly **2.48 nats/token from RETAIN**. This is why I initially considered dropping the refusal construction: it never behavior-matched the strength of actual forgetting. I retained it only as a state with clean causal provenance, since the FULL weights remain unchanged and removing the adapter restores the original model exactly.
- **The learned direction may encode refusal reversal rather than memory access.**
  Because IDK suppresses recall through a refusal policy, the $FULL - IDK$ direction may capture “answer rather than refuse” rather than a general mechanism for reopening inaccessible memories. Successful recovery therefore validates a reversible suppression axis, but not yet a memory-access axis.
- **UNLEARN and RETAIN are not behavior-matched.**
  UNLEARN starts substantially closer to FULL than RETAIN in target likelihood. Fractional recovery normalizes for this headroom difference and raw likelihood changes are reported alongside it, but the −60-60 pp differential should not be read as a comparison between behaviorally identical states.
- **The causal assay is narrow.**
  The final claim concerns one additive residual-stream direction at **layer 14**, at Q_{END}, in one model and dataset. The selected $\alpha = 1$ intervention also causes some retain-side utility movement during discovery. Failure to recover UNLEARN under this assay does not imply that other layers, token positions, subspaces, or higher-dimensional interventions would also fail.
- **The experiment does not distinguish representational reorganization from durable loss of the original pathway.**
  The UNLEARN anti-transfer is compatible with several internal changes: reorganized representations, instance-specific recovery structure, altered downstream computation, or

more substantial disruption of the original causal pathway. The experiment establishes a difference in causal recoverability, not which mechanism produced it.

Failed Experiments and Pivots

The final experiment emerged from two failed calibration attempts. Both were useful because they exposed assumptions that would otherwise have made the causal interpretation weaker.

a. Qwen and the TOFU-Alias pilot

The initial plan used **Qwen-3.5-2B** and trained every required model state locally. Because the original TOFU dataset predates Qwen-3.5, I could not safely treat raw TOFU as unseen pretraining data. I therefore constructed **TOFU-Alias**: each of the 200 fictitious authors was assigned a deterministic nonce name, replaced consistently across questions, answers, paraphrases, and perturbations, while preserving the original `full`, `forget10`, and `retain90` splits.

The intended counterfactual was:

- **FULL:** Qwen trained on all 4,000 aliased examples.
- **RETAIN:** the same initialization trained only on the 3,600 retain examples, with the forget-author bindings withheld.

Both models used the same rank-16 LoRA setup and training protocol. Five epochs were planned, but one epoch already consumed roughly **4.7 hours of compute**, making the full sweep impractical within the remaining project budget.

The pilot did demonstrate acquisition. Relative to BASE, FULL improved forget-target log-probability by **+0.8334 nats/token**, with a 95% author-clustered interval of ([0.7875, 0.8809]).

The problem was that RETAIN improved almost as much, despite never being project-trained on the forget-author bindings:

$$\mathrm{FULL} - \mathrm{RETAIN} = +0.03125 \ \mathrm{nats/token},$$

only about **3.75% of the FULL–BASE acquisition gain**. The correct-versus-perturbed margin differed by just (+0.01576).

So the run did not fail because Qwen learned nothing. It failed because it did not produce a sufficiently separated **accessible-versus-withheld comparison on the target metric**. Activation patching between those states would therefore have been difficult to interpret.

Importantly, this identifiability problem was specific to using absolute target likelihood to compare **two separately trained models**. In the final experiment, the causal quantity is instead a paired within-state difference,

$$s(M_{\text{intervened}}) - s(M),$$

measured on the same questions and in the same runtime. This removes between-model task adaptation from the causal contrast; R-control and norm-matched random directions separately test whether the intervention is causing nonspecific disruption.

One likely contributor to the Qwen failure is that aliasing author names does not make the answer distribution itself novel. TOFU answers still share vocabulary, response structure, and task regularities, so RETAIN training can improve teacher-forced target likelihood through broad task adaptation without learning the withheld author bindings themselves. More training might have widened the gap, but it was neither guaranteed to solve this identifiability problem nor affordable within the time budget.

I therefore stopped at the **pre-specified counterfactual-separation gate** and pivoted to the public **Llama-3.2-1B-Instruct TOFU checkpoint family**, which provided standardized FULL, RETAIN, and **UNLEARN** states. UNLEARN is the public checkpoint produced by applying Gradient Difference unlearning to the forget-10% subset.

b. Attempts to behavior-match a reversible IDK state

The next plan required a reversible suppression state that behaved approximately like RETAIN while leaving the original FULL model exactly recoverable. I tried two LoRA-based constructions over the frozen Llama FULL checkpoint.

IDK construction	FULL−IDK suppression	Distance to RETAIN	Additional result	Outcome
Refusal LoRA	0.1199	2.4844	retain utility passed	Failed behavioral match
Direct suppression	0.0820	2.5222	refusal margin = −0.7047	Failed behavioral match

Both constructions failed the **original** behavioral-calibration requirement: at least (0.15) nats/token of suppression and distance within (0.3) nats/token of RETAIN. Those failures remain failures. The final experiment subsequently narrowed the role of IDK: it no longer needed to imitate forgetting behaviorally, only to provide a known reversible suppression mechanism on which a recovery assay could be calibrated. The (0.1199)-nat refusal adapter satisfied that narrower role because it was removable, utility-preserving, and retained positive FULL−IDK headroom across all discovery examples.

The first adapter trained the model to produce short refusal responses on forget questions while preserving retain behavior. It was cleanly reversible: disabling the adapter restored FULL exactly, and retain-side degradation remained within the utility guardrail. But the suppression was weak and nowhere near RETAIN.

The second construction added a direct suppression objective intended to reduce correct-answer evidence more aggressively and make refusal responses preferable. It also preserved the frozen base and passed the retain-utility guardrail, but did not produce a better forgetting match.

Longer or more aggressive training may have strengthened suppression, but the deeper issue was already visible: **constructing a removable refusal policy was much easier than constructing a reversible state that behaved like a model that had never learned the targets.**

Rather than relabel either failed calibration as a success, I narrowed the role of IDK. The refusal adapter became a **known reversible suppression mechanism**, not a RETAIN-matched model of forgetting. The final experiment therefore changed from asking whether a single intervention could distinguish hidden memory from absence in behavior-matched states, to asking whether a recovery direction calibrated on one known suppression mechanism transfers differentially to UNLEARN and RETAIN.

That pivot produced the central result: the assay robustly reversed the state it was calibrated on, but anti-transferred to UNLEARN. The failed pilots therefore changed the scientific question rather than merely delaying the experiment. They exposed that **constructing a reversible suppressor is not the same as constructing a faithful model of forgetting**, and pushed the project toward studying whether causal recoverability itself depends on how forgetting was produced.